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CSE 4344-002

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Project II

Questions 1-7

<http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file1.html>

http						
No.	http	Source	Destination	Protocol	Length	Info
8114	http2	192.168.1.240	128.119.245.12	HTTP	541	GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1
953	http3	128.119.245.12	192.168.1.240	HTTP	540	HTTP/1.1 200 OK (text/html)
964		192.168.1.240	128.119.245.12	HTTP	487	GET /favicon.ico HTTP/1.1
974		128.119.245.12	192.168.1.240	HTTP	538	HTTP/1.1 404 Not Found (text/html)
1664		192.168.1.240	192.168.1.1	HTTP	324	GET /wpad.dat HTTP/1.1
1666		192.168.1.1	192.168.1.240	HTTP	203	HTTP/1.1 308 Permanent Redirect

> Frame 919: 541 bytes on wire (4328 bits), 541 bytes captured (4328 bits) on interface \Device\NPF_{693A9FD8-6A70-4DE0-8165-3E23D9359F}

> Ethernet II, Src: 54:49:43:08:12:ef (54:49:43:08:12:ef), Dst: Sagemcom_86:34:dd (f4:05:95:86:34:dd)

> Internet Protocol Version 4, Src: 192.168.1.240, Dst: 128.119.245.12

> Transmission Control Protocol, Src Port: 2091, Dst Port: 80, Seq: 1, Ack: 1, Len: 487

> Hypertext Transfer Protocol

> GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1\r\n

Host: gaia.cs.umass.edu\r\n

Connection: keep-alive\r\n

DNT: 1\r\n

Upgrade-Insecure-Requests: 1\r\n

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/111.0.0.0 Safari/537.36 Edg/11

Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0

Accept-Encoding: gzip, deflate\r\n

Accept-Language: en-US,en;q=0.9\r\n

\r\n

[Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file1.html]

[HTTP request 1/2]

[Response in frame: 953]

[Next request in frame: 964]

1. HTTP version 1.1 is the version the browser is running. The server is running Apache/2.4.6.
2. On the line "Accept-Language: ", the language accepted is en-US.

3. The IP address of my computer under “source” from the capture window is 192.168.1.240.
4. The status code returned from the capture windows under Info of No.31 was “200 OK”.

http						
No.	Time	Source	Destination	Protocol	Length	Info
5	0.006146	192.168.1.240	192.168.1.1	HTTP	273	GET /wpad.dat HTTP/1.1
7	0.010199	192.168.1.1	192.168.1.240	HTTP	203	HTTP/1.1 308 Permanent Redirect
28	0.444658	192.168.1.240	128.119.245.12	HTTP	534	GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1
31	0.518804	128.119.245.12	192.168.1.240	HTTP	540	HTTP/1.1 200 OK (text/html)
32	0.554290	192.168.1.240	128.119.245.12	HTTP	480	GET /favicon.ico HTTP/1.1
33	0.624900	128.119.245.12	192.168.1.240	HTTP	538	HTTP/1.1 404 Not Found (text/html)
1419	42.331448	192.168.1.240	192.168.1.1	HTTP	291	GET /wpad.dat HTTP/1.1
1422	42.335759	192.168.1.1	192.168.1.240	HTTP	203	HTTP/1.1 308 Permanent Redirect
1493	44.012680	192.168.1.240	192.168.1.1	HTTP	273	GET /wpad.dat HTTP/1.1
1503	44.017874	192.168.1.1	192.168.1.240	HTTP	203	HTTP/1.1 308 Permanent Redirect


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> Frame 31: 540 bytes on wire (4320 bits), 540 bytes captured (4320 bits) on interface \Device\NPF_{693A9FDB-6A70-4DE0-8165-3E23D9359F9}
> Ethernet II, Src: Sagemcom_86:34:e2 (f4:05:95:86:34:e2), Dst: 54:49:43:08:12:ef (54:49:43:08:12:ef)
> Internet Protocol Version 4, Src: 128.119.245.12, Dst: 192.168.1.240
> Transmission Control Protocol, Src Port: 80, Dst Port: 2531, Seq: 1, Ack: 481, Len: 486
▼ Hypertext Transfer Protocol
  > HTTP/1.1 200 OK\r\n
    Date: Mon, 27 Mar 2023 03:37:23 GMT\r\n
    Server: Apache/2.4.6 (CentOS) OpenSSL/1.0.2k-fips PHP/7.4.33 mod_perl/2.0.11 Perl/v5.16.3\r\n
    Last-Modified: Sun, 26 Mar 2023 05:59:01 GMT\r\n
    ETag: "80-5f7c752b0839b"\r\n
    Accept-Ranges: bytes\r\n
  > Content-Length: 128\r\n
    Keep-Alive: timeout=5, max=100\r\n
    Connection: Keep-Alive\r\n
    Content-Type: text/html; charset=UTF-8\r\n
    \r\n
    [HTTP response 1/2]
    [Time since request: 0.074146000 seconds]
    [Request in frame: 28]
    [Next request in frame: 32]
    [Next response in frame: 33]
    [Request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file1.html]
    File Data: 128 bytes
  > Line-based text data: text/html (4 lines)

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5. The file was last modified: Sun, 26 Mar 2023 05:59:01 GMT
6. 128 bytes of “File Data” were returned to my browsers.

No.	Time	Source	Destination	Protocol	Length	Info
5	0.006146	192.168.1.240	192.168.1.1	HTTP	273	GET /wpad.dat HTTP/1.1
7	0.010199	192.168.1.1	192.168.1.240	HTTP	203	HTTP/1.1 308 Permanent Redirect
28	0.444658	192.168.1.240	128.119.245.12	HTTP	534	GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1
31	0.518804	128.119.245.12	192.168.1.240	HTTP	540	HTTP/1.1 200 OK (text/html)
32	0.554290	192.168.1.240	128.119.245.12	HTTP	480	GET /favicon.ico HTTP/1.1
33	0.624900	128.119.245.12	192.168.1.240	HTTP	538	HTTP/1.1 404 Not Found (text/html)
1419	42.331448	192.168.1.240	192.168.1.1	HTTP	291	GET /wpad.dat HTTP/1.1
1422	42.335759	192.168.1.1	192.168.1.240	HTTP	203	HTTP/1.1 308 Permanent Redirect
1493	44.012680	192.168.1.240	192.168.1.1	HTTP	273	GET /wpad.dat HTTP/1.1
1503	44.017874	192.168.1.1	192.168.1.240	HTTP	203	HTTP/1.1 308 Permanent Redirect
1657	54.947356	192.168.1.240	192.168.1.1	HTTP	306	GET /wpad.dat HTTP/1.1
1659	54.951933	192.168.1.1	192.168.1.240	HTTP	203	HTTP/1.1 308 Permanent Redirect
1773	62.998095	192.168.1.240	192.168.1.1	HTTP	306	GET /wpad.dat HTTP/1.1
1775	63.002352	192.168.1.1	192.168.1.240	HTTP	203	HTTP/1.1 308 Permanent Redirect

Transmission Control Protocol, Src Port: 80, Dst Port: 2531, Seq: 1, Ack: 481, Len: 486

Hypertext Transfer Protocol

HTTP/1.1 200 OK\r\n
Date: Mon, 27 Mar 2023 03:37:23 GMT\r\n
Server: Apache/2.4.6 (CentOS) OpenSSL/1.0.2k-fips PHP/7.4.33 mod_perl/2.0.11 Perl/v5.16.3\r\n
Last-Modified: Sun, 26 Mar 2023 05:59:01 GMT\r\n
ETag: "80-5f7c752b0839b"\r\n
Accept-Ranges: bytes\r\n
Content-Length: 128\r\n
Keep-Alive: timeout=5, max=100\r\n
Connection: Keep-Alive\r\n
Content-Type: text/html; charset=UTF-8\r\n
\r\n
[HTTP response 1/2]
[Time since request: 0.074146000 seconds]
[\[Request in frame: 28\]](#)
[\[Next request in frame: 32\]](#)
[\[Next response in frame: 33\]](#)
[Request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file1.html]
File Data: 128 bytes

Line-based text data: text/html (4 lines)

<html>\n
Congratulations. You've downloaded the file \n
http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file1.html!\n
</html>\n

7. No, I don't see any headers from the raw data that aren't displayed in the packing window.

Questions 8 – 11

<http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file2.html>

No.	Time	Source	Destination	Protocol	Length	Info
22	10.291988	192.168.1.240	192.168.1.1	HTTP	306	GET /upad.dat HTTP/1.1
24	10.296534	192.168.1.1	192.168.1.240	HTTP	203	HTTP/1.1 308 Permanent Redirect
47	10.451543	192.168.1.240	128.119.245.12	HTTP	534	GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
49	10.524828	128.119.245.12	192.168.1.240	HTTP	784	HTTP/1.1 200 OK (text/html)
74	19.021018	192.168.1.240	128.119.245.12	HTTP	646	GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
77	19.095127	128.119.245.12	192.168.1.240	HTTP	294	HTTP/1.1 304 Not Modified

> Frame 47: 534 bytes on wire (4272 bits), 534 bytes captured (4272 bits) on interface \Device\NPF_{693A9FDB-6A70-4DE0-8165-3E23D9359}

> Ethernet II, Src: 54:49:43:08:12:ef (54:49:43:08:12:ef), Dst: Sagemcom_86:34:dd (f4:05:95:86:34:dd)

> Internet Protocol Version 4, Src: 192.168.1.240, Dst: 128.119.245.12

> Transmission Control Protocol, Src Port: 2676, Dst Port: 80, Seq: 1, Ack: 1, Len: 480

Hypertext Transfer Protocol

GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1\r\n

[Expert Info (Chat/Sequence): GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1\r\n]

[GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1\r\n]

[Severity level: Chat]

[Group: Sequence]

Request Method: GET

Request URI: /wireshark-labs/HTTP-wireshark-file2.html

Request Version: HTTP/1.1

Host: gaia.cs.umass.edu\r\n

Connection: keep-alive\r\n

DNT: 1\r\n

Upgrade-Insecure-Requests: 1\r\n

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/111.0.0.0 Safari/537.36\r\n

Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exch

Accept-Encoding: gzip, deflate\r\n

Accept-Language: en-US,en;q=0.9\r\n

\r\n

[Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file2.html]

[HTTP request 1/1]

[Response in frame: 49]

8. No, I didn't see anything that said "IF-MODIFIED-SINCE" in the first HTTP GET.

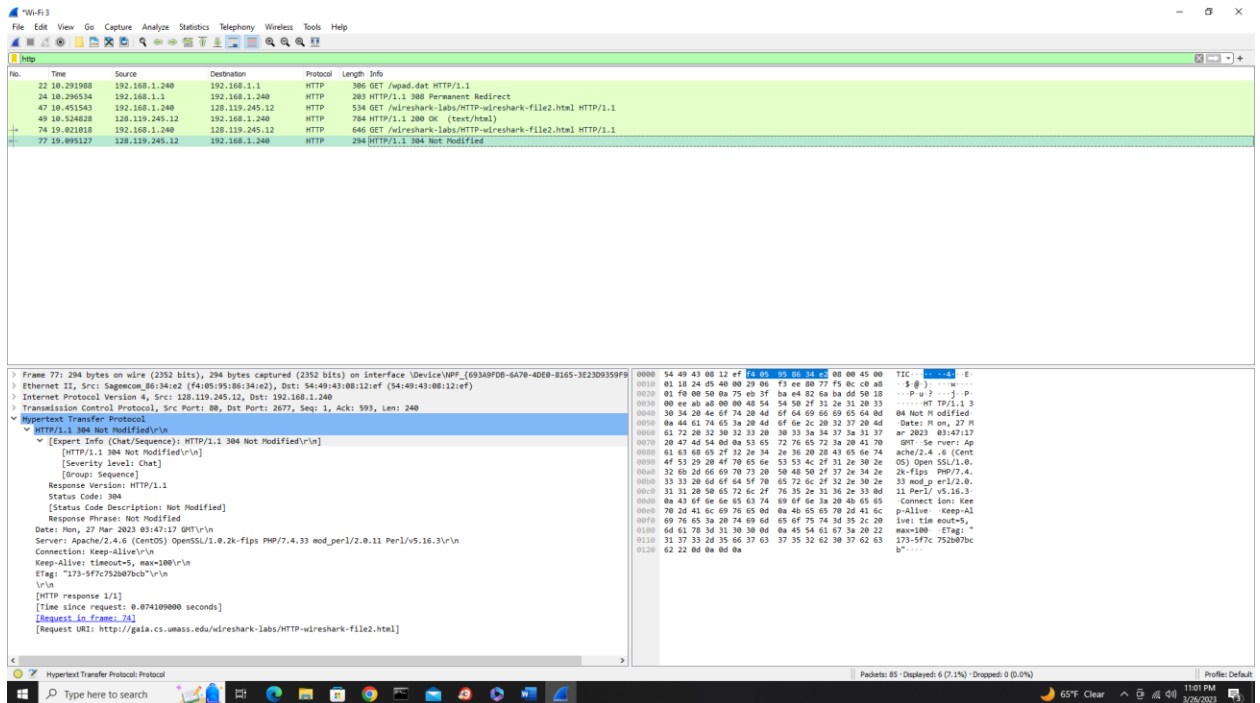
```

Hypertext Transfer Protocol
  HTTP/1.1 200 OK\r\n
    [Expert Info (Chat/Sequence): HTTP/1.1 200 OK\r\n]
      [HTTP/1.1 200 OK\r\n]
      [Severity level: Chat]
      [Group: Sequence]
      Response Version: HTTP/1.1
      Status Code: 200
      [Status Code Description: OK]
      Response Phrase: OK
      Date: Mon, 27 Mar 2023 03:47:08 GMT\r\n
      Server: Apache/2.4.6 (CentOS) OpenSSL/1.0.2k-fips PHP/7.4.33 mod_perl/2.0.11 Perl/v5.16.3\r\n
      Last-Modified: Sun, 26 Mar 2023 05:59:01 GMT\r\n
      ETag: "173-5f7c752b07bcb"\r\n
      Accept-Ranges: bytes\r\n
    > Content-Length: 371\r\n
      Keep-Alive: timeout=5, max=100\r\n
      Connection: Keep-Alive\r\n
      Content-Type: text/html; charset=UTF-8\r\n
      \r\n
      [HTTP response 1/1]
      [Time since request: 0.073285000 seconds]
      [Request in frame: 47]
      [Request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file2.html]
      File Data: 371 bytes
  Line-based text data: text/html (10 lines)

```

[illegible]

10. Yes, and the line says “If-Modified-Since: Sun, 26 Mar 2023 05:59:01 GMT”.



11. The HTTP status code is “304 Not Modified”. The server didn’t return the content of the file as there was no data returned. By checking the heading date, it was understood that no data was changed on the site and so it returned the same information.

Questions 12 – 15

<http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file3.html>

Wireshark interface showing network traffic analysis. The main pane displays a list of captured packets, with packet 70 selected. The packet details pane shows the structure of the selected packet, including Ethernet II, Internet Protocol Version 4, and Hypertext Transfer Protocol. The packet bytes pane displays the raw data of the selected packet.

The packet list shows the following details for packet 70:

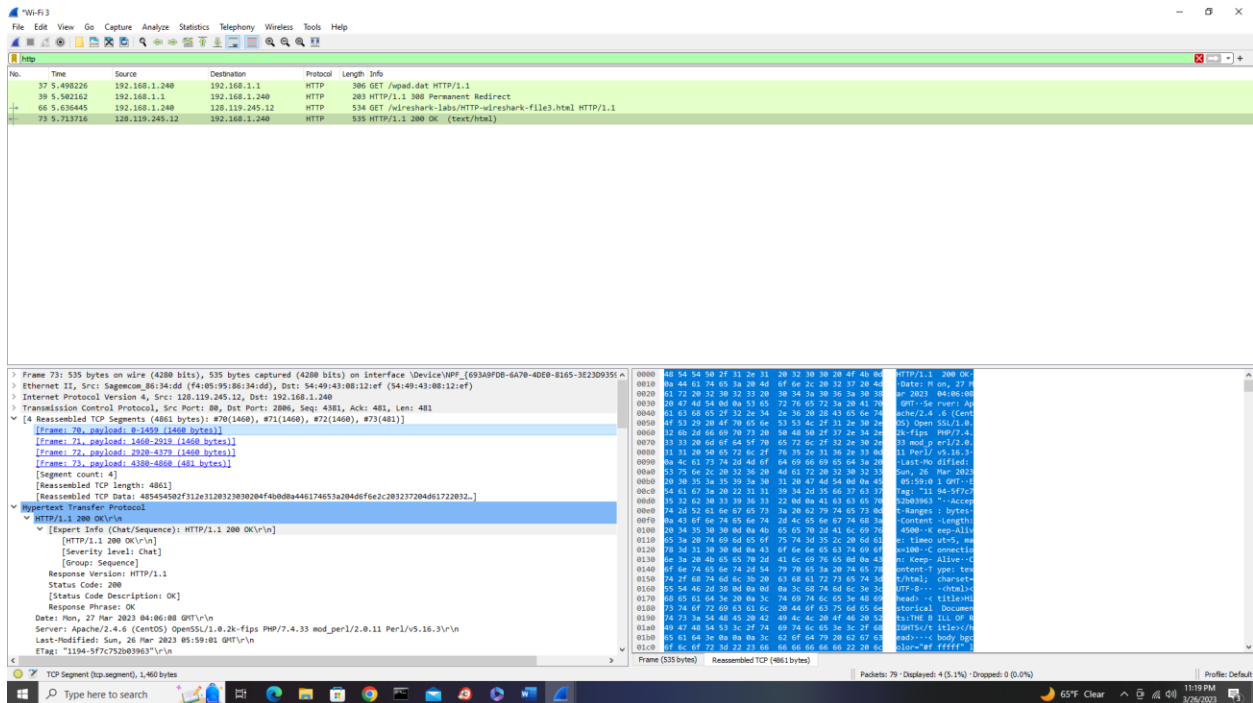
No.	Time	Source	Destination	Protocol	Length	Info
70	37.5408226	192.168.1.240	192.168.1.1	HTTP	306	GET /wpad.dat HTTP/1.1

The packet details pane shows the following structure for packet 70:

- Ethernet II, Src: 54:00:43:00:12:1f (54:00:43:00:12:1f), Dst: 08:00:2b:01:00:00 (08:00:2b:01:00:00)
- Internet Protocol Version 4, Src: 192.168.1.240, Dst: 128.119.245.12
- Hypertext Transfer Protocol
 - [Expert Info (Chat/Sequence): GET /wpad.dat HTTP/1.1/v/n]
 - [Severity Level: Chat]
 - [Group: Sequence]
 - Request Method: GET
 - Request URI: /wpad.dat
 - Request Version: HTTP/1.1
 - Host: 128.119.245.12
 - Connection: keep-alive
 - DNT: 1
 - User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/111.0.0.0 Safari/537.36
 - Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
 - Accept-Encoding: gzip, deflate
 - Accept-Language: en-US,en;q=0.9
 - [Full request URI: http://128.119.245.12/wpad.dat]
 - [HTTP request 1/1]
 - [Response in frame 71]

The packet bytes pane displays the raw data of the selected packet, showing the structure of the packet and the raw data of the selected packet.

12. Only one request message was sent, so the packet number in the trace was packet number 70.



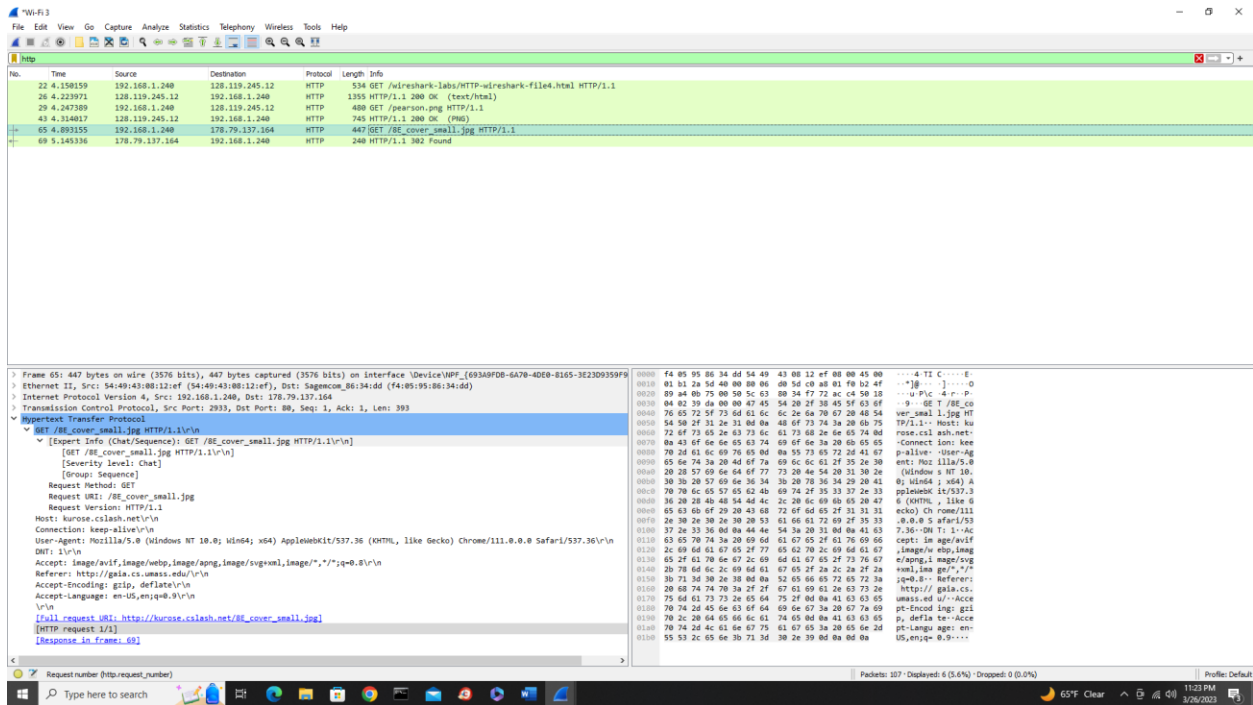
13. The packet with the response is the beginning packet, packet 70.

14. The status code and phrase are 200 OK.

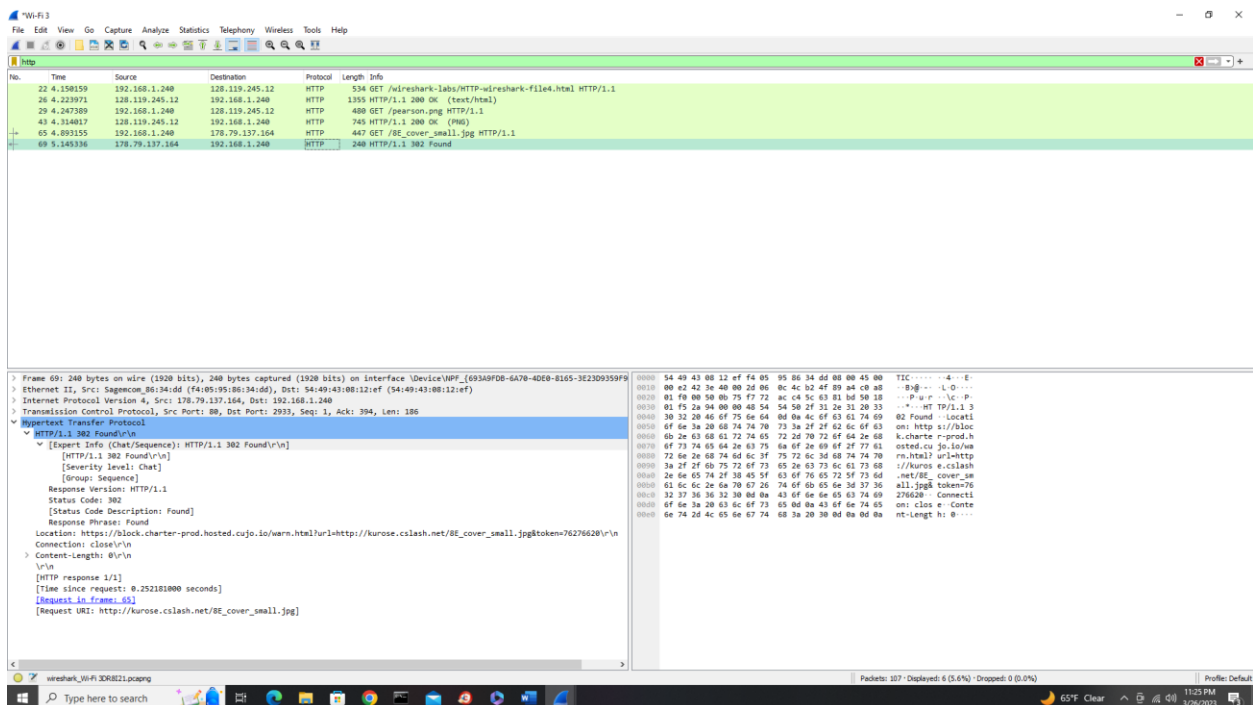
15. There were four TCP segments needed to carry the single TTP response and the text of the Bill of Rights.

Questions 16 – 17

<http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file4.html>



16. There were three GET request messages that my browser sent. The internet addresses for the first two were the same, but the third was different (128.119.245.12, 178.79.137.164).



17. It can't exactly be told if the two images were downloaded serially or not, because there's no dating for the second image. Instead, it has a time since the request, but not a time for when this image was requested to compare the first image.

Questions 18 – 19

<http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file5.html>

The image shows a Wireshark packet capture of an HTTP transaction. The packet list on the left shows several packets, with packet 42 being the GET request and packet 43 being the 200 OK response. The packet details pane for packet 43 is expanded, showing the HTTP response structure. The response is a 200 OK status code, indicating a successful GET request. The response body is a JPEG image, as indicated by the 'Content-Type: image/jpeg' header. The packet bytes pane shows the raw data of the response, including the status bar and the image data.

No.	Time	Source	Destination	Protocol	Length	Info
40	2.054252	192.168.1.240	128.119.245.12	HTTP	534	GET /wireshark-labs/HTTP-wireshark-file5.html HTTP/1.1
42	2.127597	128.119.245.12	192.168.1.240	HTTP	887	HTTP/1.1 200 OK (text/html)
44	2.160161	192.168.1.240	128.119.245.12	HTTP	494	GET /wireshark-labs/cover.jpg HTTP/1.1
71	2.239948	192.168.1.240	128.119.245.12	HTTP	447	GET /kurose/cover.jpg HTTP/1.1
72	2.315094	128.119.245.12	192.168.1.240	HTTP	1514	[TCP Previous segment not captured] Continuation
73	2.315094	128.119.245.12	192.168.1.240	HTTP	1514	Continuation
76	2.315094	128.119.245.12	192.168.1.240	HTTP	1514	Continuation
77	2.315094	128.119.245.12	192.168.1.240	HTTP	1514	Continuation
78	2.315094	128.119.245.12	192.168.1.240	HTTP	1514	Continuation
79	2.315094	128.119.245.12	192.168.1.240	HTTP	1514	Continuation
91	2.392027	128.119.245.12	192.168.1.240	HTTP	1514	Continuation
92	2.392027	128.119.245.12	192.168.1.240	HTTP	1514	Continuation
93	2.392027	128.119.245.12	192.168.1.240	HTTP	1514	Continuation
99	2.494286	128.119.245.12	192.168.1.240	HTTP	480	HTTP/1.1 200 OK (JPEG image)
103	2.735872	128.119.245.12	192.168.1.240	HTTP	1514	Continuation
104	2.735872	128.119.245.12	192.168.1.240	HTTP	1514	Continuation
105	2.735872	128.119.245.12	192.168.1.240	HTTP	1514	Continuation
106	2.735872	128.119.245.12	192.168.1.240	HTTP	1514	Continuation
107	2.735872	128.119.245.12	192.168.1.240	HTTP	1514	Continuation
108	2.735872	128.119.245.12	192.168.1.240	HTTP	1514	Continuation
109	2.735872	128.119.245.12	192.168.1.240	HTTP	1514	Continuation
110	2.735872	128.119.245.12	192.168.1.240	HTTP	1514	Continuation
112	2.804689	128.119.245.12	192.168.1.240	HTTP	1514	Continuation
113	2.804689	128.119.245.12	192.168.1.240	HTTP	1514	Continuation
114	2.804689	128.119.245.12	192.168.1.240	HTTP	1514	Continuation
115	2.804689	128.119.245.12	192.168.1.240	HTTP	1514	Continuation

Frame 40: 534 bytes on wire (4272 bits), 534 bytes captured (4272 bits) on Interface \Device\NPF_{693A9F08-6A70-40E0-B165-3E230935E} Ethernet II, Src: 54-89-43-08-12-ef (54:89:43:08:12:ef), Dst: Sagemcom_06:34:dd (f4:85:95:06:34:dd)

Internet Protocol Version 4, Src: 192.168.1.240, Dst: 128.119.245.12

Transmission Control Protocol, Src Port: 3819, Dst Port: 80, Seq: 1, Ack: 1, Len: 480

Hypertext Transfer Protocol

GET /wireshark-labs/HTTP-wireshark-file5.html HTTP/1.1

[Expert Info (Chat/Sequence): GET /wireshark-labs/HTTP-wireshark-file5.html HTTP/1.1/v/n]

[GET /wireshark-labs/HTTP-wireshark-file5.html HTTP/1.1/v/n]

[Severity level: Chat]

[Group: Sequence]

Request Method: GET

Request URI: /wireshark-labs/HTTP-wireshark-file5.html

Request Version: HTTP/1.1

Host: gaia.cs.umass.edu/v/n

Connection: keep-alive/v/n

DNT: 1/v/n

Upgrade-Insecure-Requests: 1/v/n

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/110.0.0.0 Safari/537.36/v/n

Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.9/v/n

Accept-Encoding: gzip, deflate/v/n

Accept-Language: en-US,en;q=0.9/v/n

[Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file5.html]

[HTTP request 1/2]

[Response in frame 42]

[Next request in frame 44]

18. The initial response from the get is 200 OK on No. 42.

19. The new response from the second is a continuation with no status code.